TREKVERSE

Intelligent Academic Performance Monitoring & Recommendation System

Recommended Project Upgrade Proposal

Project Focus	Academic performance monitoring, predictive analytics, early identification of learning needs, and personalized recommendations.
Core Users	Administrator, faculty, and student
Suggested Stack	Python, Streamlit, scikit-learn, SHAP, SQL database, data visualization, and PDF reporting

Purpose: evolve a basic student-grade prediction project into a practical, explainable, and data-backed academic monitoring platform. Predictions should support human review and student guidance—not replace faculty judgment.

Prepared from the supplied project-upgrade requirements, adapted to the selected project title.

1. Executive Summary

The proposed upgrade transforms the existing student-performance prediction application into the **Intelligent Academic Performance Monitoring & Recommendation System**. Instead of predicting only a final grade, the platform combines performance forecasting, academic progress monitoring, risk indicators, explainable predictions, student engagement analysis, personalized study guidance, role-based dashboards, and downloadable reports.

The project can retain its existing Streamlit and machine-learning foundations while adding a modular architecture, database persistence, college-data import, and stronger model validation.

Recommended target capabilities

- **Monitor:** track attendance, assessments, engagement, subject-level performance, and semester trends.
- **Predict:** estimate student performance and identify students who may need academic support.
- **Explain:** show the factors that influenced each prediction through SHAP or another suitable explanation method.
- **Recommend:** generate actionable study recommendations and faculty intervention suggestions.
- **Manage:** provide separate administrator, faculty, and student experiences with appropriate access controls.
- **Report:** export student-level academic summaries as professional PDF reports.

Important: all example dashboard figures and student scores in this proposal are illustrative. They must not be displayed as live results unless generated from validated project data.

2. Current Project vs. Proposed Upgrade

Current capability	Proposed modification
Final-grade prediction	Performance forecasting plus an early-warning risk estimate
Single student input	Student profile, login, and academic history
Random Forest model	Compare suitable models using consistent validation
SHAP explanation	Prediction-level explanation with clear limitations
Rule-based recommendations	Personalized study plan based on weak areas and student goals
K-Means clustering	Engagement and behavior segments, interpreted cautiously
Single Streamlit screen	Admin, faculty, student, analytics, and reporting pages
UCI dataset only	Validated CSV/Excel import for college data
Basic prediction	What-if scenario simulator
Basic warnings	Configurable early-warning alerts and follow-up workflow
No report workflow	Downloadable PDF student report
Static CSV files	Database-backed student, assessment, and prediction records

3. Academic Monitoring & Early-Warning System

The monitoring feature should combine multiple indicators to flag students who may benefit from timely academic support. The initial design can use transparent thresholds, a validated risk model, or a hybrid approach. Thresholds should be agreed with the institution and checked for false alarms and missed cases.

Risk bands

Level	Meaning	Suggested follow-up
Low Risk	Current indicators do not show an elevated concern under the configured rules/model.	Continue normal monitoring and positive feedback.
Moderate Risk	One or more indicators suggest that a check-in or targeted support may be useful.	Faculty check-in, review weak subjects, and agree on a short action plan.
High Risk	Multiple or high-priority indicators suggest a need for prompt human review.	Contact the student, confirm context, and coordinate suitable academic support.

Potential input factors

- Attendance percentage and attendance trend
- Previous semester CGPA and prior failures/backlogs
- Internal marks, quiz performance, and subject-level assessment scores
- Assignment scores and submission rate
- Reported study time, where collected consistently and with consent
- Learning-platform activity, using only appropriate and clearly defined measures

Illustrative dashboard

Students Requiring Support	Moderate Risk	Low Risk
24 (example)	57 (example)	214 (example)

These figures are placeholders, not verified student counts. The live dashboard must calculate counts from the current dataset and display the reporting date.

4. What-If Prediction Simulator

Allow students or faculty to adjust selected inputs and compare the model's baseline prediction with a simulated scenario. The application should explain which inputs changed and should keep the original student record unchanged unless an authorized user explicitly saves an update.

Input	Current example	Scenario example
Attendance	72%	90%
Study time	2 hours/day	4 hours/day
Assignment completion	60%	95%
Predicted score	12.8 / 20	15.4 / 20

The scores above are illustrative only. A simulator must use the trained model to calculate outputs; it must not hard-code the example result. Label outputs as estimates, not guarantees, and avoid suggesting unrealistic changes outside the model's training range.

5. Role-Based Dashboards

Student Dashboard

- Profile and academic history
- Current CGPA, attendance, and subject performance
- Semester trends and predicted performance
- Risk level with an explanation of contributing factors
- Strong subjects, subjects needing attention, and personalized recommendations
- What-if simulator and downloadable report

Faculty Dashboard

- Total students, class average, attendance average, and risk distribution
- Top performers and students who may need support
- Subject analysis, student comparison, and performance trends
- Student detail view with marks, attendance, previous failures, and explanation
- Intervention recommendations, follow-up status, and faculty remarks

Example student view (illustrative)

Field	Example value
Student ID	STU1024
Predicted score	13.6 / 20
Risk band	Moderate
Attendance	68%
Potential focus area	Mathematics
Previous failures	1
Suggested next step	Additional practice and an attendance check-in

Administrator Dashboard

- Manage users, departments, semesters, subjects, and role permissions
- Import datasets and review validation errors
- Monitor model versions, prediction runs, data quality, and system activity
- Configure risk thresholds and report settings with appropriate governance

6. Subject-Wise Prediction and Analytics

Extend the prediction view from one overall grade to subject-level estimates where the available data supports them. Use the same score scale for all subjects or clearly display the applicable scale.

Subject	Example predicted score	Interpretation
Mathematics	15.2	Review alongside past assessments
Programming	17.1	Potential strength
DBMS	13.8	May need targeted practice
Computer Networks	14.6	Monitor progress
Electronics	12.9	May need targeted practice

The subject scores are illustrative. Strong/weak labels should be based on defined academic criteria and uncertainty, not arbitrary comparisons alone.

Suggested visual analytics

- Semester-wise CGPA and marks trend
- Subject performance comparison
- Attendance versus assessment performance
- Risk-band distribution and changes over time
- Prediction error distribution and model performance indicators

7. Machine-Learning Pipeline

The existing project may report a tuned Random Forest model with an R^2 around 0.285. Treat that as a reported baseline that should be re-tested using the exact dataset, feature definitions, split strategy, and code version. A low or moderate score does not automatically make the project unusable, but it does mean the model should be presented with appropriate uncertainty and limitations.

Recommended training workflow

```
Raw data
  ↓
Schema checks + missing-value handling
  ↓
Feature engineering + leakage review
  ↓
Train / validation / test split
  ↓
Baseline: mean predictor + Linear Regression
  ↓
Random Forest / Gradient Boosting
  ↓
Optional: XGBoost / LightGBM / CatBoost
  ↓
Cross-validation + hyperparameter tuning
  ↓
Final holdout evaluation + error analysis
  ↓
Versioned model + documented limitations
```

Compare candidate models only when the dependencies are available and the evaluation setup is identical. Include a simple baseline, and choose the final model based on appropriate validation—not on training performance alone.

Metric	What it communicates
MAE	Average absolute prediction error in the target's units
RMSE	Penalizes larger prediction errors more heavily
R^2	How much target variation is explained relative to a mean baseline
Cross-validation	How performance varies across training folds
Residual/error analysis	Where and how predictions are inaccurate

Model and fairness safeguards

- Prevent data leakage: do not train on information that would not be available at prediction time.
- Keep a final holdout set separate from model tuning.
- Report uncertainty and subgroup performance where sample sizes support meaningful analysis.
- Document feature definitions, model version, training date, and dataset source.
- Use SHAP as an explanation aid, not proof of causation; explanations can be unstable or misleading for correlated features.
- Have faculty review risk flags before consequential decisions. Do not use the model as the sole basis for penalties, admissions, or progression decisions.

8. Dataset Import and Validation

Support institutional data uploads in CSV and Excel (.xlsx) formats. A mapping step should let the user match incoming column names to the application's expected fields, rather than assuming every college uses identical headers.

Suggested field	Validation example
Student_ID	Required; unique within the agreed scope

Suggested field	Validation example
Name	Optional for model training; restrict access
Department / Semester	Allowed values or institution-defined mapping
Attendance	Numeric percentage within 0–100
Internal_Marks / Assignment_Marks / Quiz_Marks	Numeric values within configured assessment ranges
Study_Hours	Non-negative; flag implausible values for review
Previous_CGPA	Within institution's configured CGPA scale
Backlogs	Non-negative integer
Final_Marks	Target field; ensure it is not accidentally used as an input for a prediction made before the final exam

Upload workflow

- Preview the file and detected columns before importing.
- Check required fields, data types, allowed ranges, duplicates, and missing values.
- Show row-level errors and allow a corrected file to be uploaded.
- Confirm consent, authorization, and permitted purpose before processing student data.
- Keep raw uploads access-controlled; avoid logging names, marks, or other personal data unnecessarily.

9. Database and Application Architecture

```
Admin / Faculty / Student UI
          ↓
Streamlit application (or web frontend)
          ↓
Authentication + role authorization
          ↓
Application / prediction service
          ↓
MySQL or PostgreSQL database
          ↓
Versioned ML model + preprocessing pipeline
          ↓
PDF report generator
```

Suggested database entities

Table	Purpose
users	Login identity, role, and account status
students	Student profile and institution identifiers
subjects	Subject catalog and semester mapping
attendance	Attendance records by student and date/course
marks	Internal, quiz, assignment, and exam results
assignments	Assignment metadata and submission status
predictions	Predictions, timestamp, model version, and target
recommendations	Suggested actions and status/history
interventions	Faculty follow-up notes and outcome tracking

Security requirements

- Enforce server-side role checks; hiding a page is not sufficient authorization.
- Students can access only their own records; faculty access should be limited to assigned students/classes.
- Use secure password hashing or a trusted identity provider, session controls, and least-privilege database credentials.
- Protect exports, audit sensitive access, and define data retention/deletion rules with the institution.
- Use environment variables or a secrets manager for credentials; never commit secrets into source control.

10. Automatic PDF Student Report

Create a consistent, downloadable report for a selected student and semester. The report should include the generation date, data period, model version, and a clear note that predictions are estimates.

Report section	Content
Header	TREKVERSE — Academic Performance Monitoring Report
Student information	Student ID, name (if authorized), department, semester
Academic performance	CGPA, attendance, average marks, subject-wise results
AI analysis	Predicted score, risk band, factors influencing prediction
Study support	Strengths, areas for practice, personalized recommendations
Faculty remarks	Optional authorized comments and follow-up plan
Disclaimer	Model output is an estimate and should be reviewed with actual academic context

11. AI-Personalized Study Planner

Use student goals, upcoming assessments, subject weaknesses, and available study time to produce a practical weekly plan. The first version can use transparent rules and templates; a generative AI component may later personalize the wording, but should not invent marks, deadlines, or course requirements.

Day	Illustrative activity
Monday	DBMS concepts — 1.5 hours
Tuesday	SQL practice — 1 hour
Wednesday	DBMS revision — 1 hour
Thursday	Previous questions — 1.5 hours
Friday	Mock test — 45 minutes

Personalization inputs may include weak topics, target grade, time available, preferred learning format, and assessment dates. Let the student edit the plan and mark tasks complete.

12. UI Design Direction

Use a modern education-platform interface with consistent typography, accessible contrast, clear navigation, and concise dashboard cards. Make the same data definitions and reporting periods consistent across pages.

Dashboard card	Example display
Total Students	296 (example only)
Average CGPA	7.84 (example only)
Students Requiring Support	24 (example only)
Average Attendance	86% (example only)

Recommended navigation

- Overview and performance analytics
- Student directory and profile
- Prediction and what-if simulator
- Early-warning queue and intervention tracking
- Data import and quality checks
- Study recommendations and reports
- User management and system settings

13. Suggested Project Structure

```
Intelligent-Academic-Monitoring/
■■■ app/
■   ■■■ app.py
■   ■■■ pages/
■       ■■■ admin.py
■       ■■■ faculty.py
■       ■■■ student.py
■       ■■■ analytics.py
■       ■■■ prediction.py
■       ■■■ reports.py
■■■ data/
■   ■■■ raw/
■   ■■■ processed/
■   ■■■ uploads/
■■■ models/
■   ■■■ model_registry/
■   ■■■ preprocessor/
■■■ src/
■   ■■■ preprocessing/
■   ■■■ feature_engineering/
■   ■■■ training/
■   ■■■ prediction/
■   ■■■ explainability/
■   ■■■ recommendations/
■   ■■■ clustering/
■■■ database/
■   ■■■ schema.sql
■   ■■■ database.py
■■■ reports/
■   ■■■ report_generator.py
■■■ tests/
■■■ requirements.txt
■■■ README.md
```

Store trained models and preprocessing artifacts as versioned build outputs; avoid committing sensitive student data or production database files to the repository.

14. Recommended Implementation Roadmap

Phase	Deliverables	Acceptance check
1. Foundation	Repository review, data dictionary, baseline model, data-quality checks	Reproducible training and documented baseline metrics
2. Prediction	Model comparison, cross-validation, holdout testing, SHAP explanations	Metrics and explanations available for a saved model version
3. Risk monitoring	Risk bands, configurable thresholds, risk dashboard	Counts derive from actual data; each flag has an explanation
4. Student/faculty UI	Role-based pages, subject analytics, what-if simulator	Role access tests pass; scenarios are not saved accidentally
5. Data and database	CSV/XLSX import, schema validation, SQL persistence	Invalid files rejected with actionable error messages
6. Recommendations/reports	Study planner, PDF report, faculty intervention workflow	Report output validated; recommendation history is tracked
7. Quality and release	Security checks, test suite, documentation, deployment	No known critical access-control issues; setup is reproducible

15. Success Criteria

- Model performance is measured on held-out data and compared with a simple baseline.
- Risk alerts are transparent, configurable, and reviewed by an authorized person.
- Uploaded files receive clear validation feedback and cannot silently corrupt records.
- Students and faculty see only data they are authorized to access.
- What-if results are generated by the model and labeled as estimates.
- PDF reports contain correct student/semester data and explain the limits of predictions.
- The project can be installed and run from documented setup instructions.

16. Final Recommendation

Build the upgrade as a series of testable modules rather than adding every feature at once. Start with data quality and reliable model evaluation, then add risk monitoring and explainability, followed by role-based dashboards, database-backed records, study planning, and PDF reports.

The proposed end product is **Intelligent Academic Performance Monitoring & Recommendation System**—an explainable academic-support platform that helps students understand their progress and helps faculty identify where timely support may be useful.

Responsible-use note: student predictions can be wrong and may reflect incomplete or biased data. Use them as decision support, provide a way to correct records or challenge a flag, and keep consequential decisions under qualified human review.